A. Manoj Reddy
192311171

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: **Theory of Computation** **COURSE CODE:** **CSA13**

COURSE OUTCOME COVERED

CO2: **Deign Finite Automata DFA, NFA, ε-NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)**

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:25

Design Task

Code of Conduct:

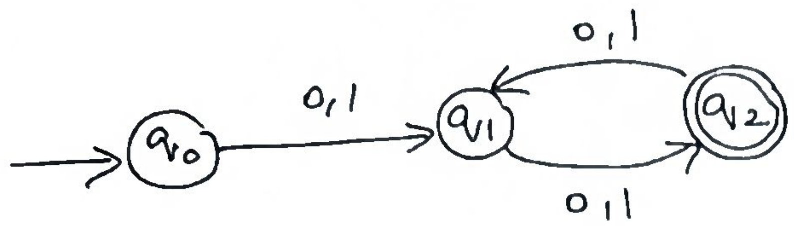
I A. Manoj Reddy 192311171 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Manoj

SIMATS ENGINEERING
ASSESSMENT TOOL

1.Design the regular expression for the language represented by the given automata



2.Consider the following grammar

S->aSc|T|U|e

T-> bTc|e

U->aUb|e

A} What is the language defined by the grammar?

B)Give the derivation tree for the word aaabbc

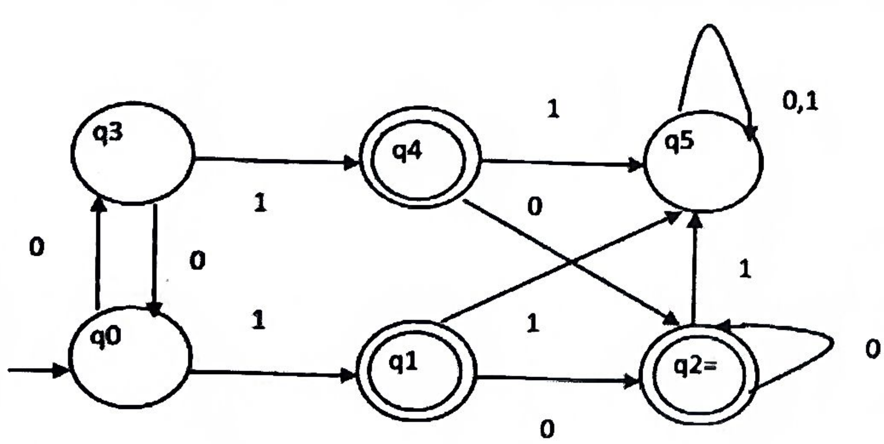
C)A Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

3. Design a non deterministic finite automata with ε-transitions to represent the language denoted by the regular expression
00*1+11*0

4. I)Show that if L1 and L2 are regular then (L1+l2) also regular

Ii)Is the language { a^p| p is prime} regular? Justify

5. Design minimized automata, Consider the DFA given below. Find the states that are equivalent and construct a minimum state automata equivalent to the given automata. Draw the minimized automata, write its transition table and mark initial and final states



Regular Expresion for given Automata

From the given automata the first 2 symbols can either '0' or '1' and automata return between q_1 and q_2

Transition table:-

State	0	1
$\rightarrow q_0$	q_1	q_1
q_1	q_2	q_2
q_2	q_1	q_1

Regular expresion

$$((0+1)(0+1))^* \; ((0+1)(0+1))^*$$

$$\Rightarrow ((0+1)(0+1))^*$$

State diagram:-

Example:-

→ 01 ✓

→ 00 ✓

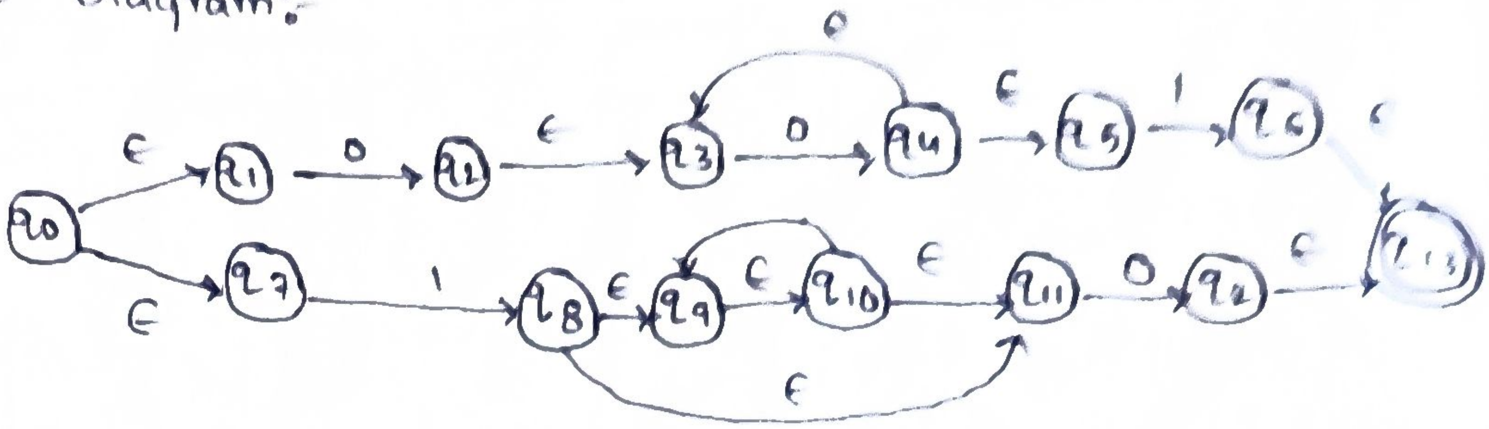
→ 0 ×

→ 101 ×

2) Given RE

$$00^*1 + 11^*0$$

State Diagram:-



Transition tables-

State	0	1	ϵ
$\rightarrow q_0$	—	—	$\{q_1, q_7\}$
q_1	q_2	—	-
q_2	—	—	$\{q_3, q_5\}$
q_3	q_4	—	—
q_4	—	—	q_3
q_5	—	—	q_{13}
q_6	—	q_8	—
q_7	—	—	q_9
q_8	—	q_{10}	—
q_9	—	—	q_{11}
q_{10}	—	—	—
q_{11}	q_{12}	—	—

② Given

$S \rightarrow aSc \mid T \mid U \mid \epsilon$

$T \rightarrow bTc \mid \epsilon$

$U \rightarrow aUb \mid \epsilon$

(ii) Is $L = \{ a^b \mid p \text{ is prime} \}$ regular?

No, the language is not regular

Assume L is regular and m be its pumping

Choose prime $p > m$

$w = a^p$

by pumping lemma, let $w = xyz$

where $y = a$, $p > 0$

$p + pk = p(1+k)$, $p(1+k)$ is composite

$\therefore \{ a^p \mid$ is p which is prime $\}$ not regular

5) from the given DFA:- final states are

q_1, q_2, q_4

Non final: q_0, q_3, q_5

State transition table:-

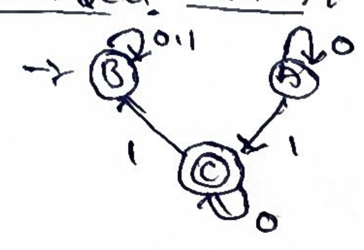
State / IP	0	1
$\rightarrow q_0$	q_3	q_1
q_1	q_2	q_5
q_2	q_2	q_4
q_3	q_0	q_5
q_4	q_3	q_5
q_5	q_5	q_5

P_0: $\{q_0\ q_3 q_5\}\ \{q_1\ q_2\ q_7\}$

P_1: $\{q_5\}\ \{q_2 q_5\}\ \{q_1\ q_2 q_4\}$

P_2: $\{q_5\}\ \{q_0\ q_3\}\ \{q_1\ q_2\ q_4\}$

Minimized DFA:-



(a) language definition:-

- T generates $b^m c^m$
- V generates $a^m b^m$
- S add matching a and c

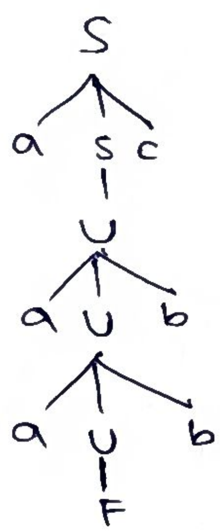
$$L = \{ a^i b^j c^{i+j} \} \cup \{ a^{i+j} b^j c^i \}$$

(b) Derivation tree for aaabbc:-

$S \to aSc$ $S \to U$ $U \to aUb$

$S \to aaUbc$ $U \to aUb$

$S \to aaabbc$ $U \to \epsilon$



(c) ambigous

$S \Rightarrow \epsilon$

$S \Rightarrow T \Rightarrow \epsilon$

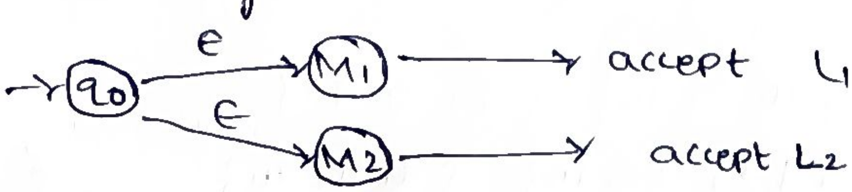
$S \Rightarrow U \Rightarrow \epsilon$

$\therefore$ Grammar is Ambigous.

4) Regular language:-

(i) Show that if L_1 and L_2 regular, $L_1 + L_2$ regular

Since L_1 and L_2 are regular, the finite Automata M_1 and M_2 accepting them amr Nuct NFA



$\therefore$ New NFA Accepts: $L_1 \cup L_2$

$\therefore L_1 \cup L_2$ is regular.

SIMATS ENGINEERING
ASSESSMENT TOOLS

:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

88/100